

Replication of an adaptive recloser–fuse protection scheme on the IEEE 13-node feeder in DigSILENT PowerFactory

Reference paper: M. Yousaf, A. Jalilian, K. M. Muttaqi, D. Sutanto, *An Adaptive Overcurrent Protection Scheme for Dual-Setting Directional Recloser and Fuse Coordination in Unbalanced Distribution Networks With Distributed Generation*, IEEE Trans. Ind. Appl., vol. 58, no. 2, pp. 1831–1842, 2022.

Tool: DigSILENT PowerFactory 2021 SP2 (Python scripting). Date: 30 September 2026. All results below come from the PowerFactory model: load flow and short-circuit (Complete method) results, and the operating times of the protective devices computed from those currents.

Status at a glance

Paper item	Status	Result file / script
Table I – DG short-circuit data	In the model; not yet re-checked	IEEE 13 Node Feeder.pfd
Table II – rated / min / max branch currents	Done	Result\Table_II.csv (IEEE13_Tables.py)
Table III – fuse coefficients b_i	Done	Result\Table_III.csv (IEEE13_Tables.py)
Table IV – R1, R2, fuse operating times	Done	Result\Table_IV_model.csv (Table_IV.py)
Fig. 6 – feeder with protective devices	Reclosers and 15 fuses placed; diagram not redrawn	Add_Fuses.py, Configure_Reclosers.py
Fig. 7 – CTI vs DG penetration	Done	Figures\Fig07
Figs. 8, 9, 11–13, 15, 16 – operation sequences	Done	Figures\Fig08 ... Fig16
Fig. 10 – time-domain fault current	Not started	needs RMS/EMT simulation
Figs. 14, 17 – coordination maps	Done	Figures\Fig14, Fig17 (Fig14_Coordination.py)
Figs. 18–21, Table V – IEEE 34-node	Out of scope so far	

Key findings

- The feeder model reproduces the paper's load flow (rated currents within 1 % of Table II on every branch) and most fault currents within 20 %; $I_{f,max}$ differs by 28–46 % on four branches (section 2).
- Recloser R1 (GE IAC77B801A extremely inverse, pickup 720 A, TDS 0.5 / 10) reproduces every R1 time given in the paper within 1–4 %.
- The coordination maps reproduce the paper's main result: without the dual-setting directional recloser (DSDR) coordination is lost for faults upstream of R2; with the DSDR it is restored. Fig. 14: 23 of 31 comparable cells agree; Fig. 17: 29 of 31.
- Fuse and R2 times differ from the paper's figures. The paper's plotted fuse times cannot be obtained from its own eq. (6) with its Table III b_i ; the model follows the equations (decision taken with the method, not the plots).

1. Model and protection settings

IEEE 13-node test feeder (4.16 kV, 60 Hz) with a 4.05 MVA / 0.69 kV synchronous DG connected at node 692 through a 0.69/4.16 kV transformer (Table I data). Short circuits use the Complete method; the IEC/ANSI methods cannot represent the feeder's single-phase regulators and one- and two-phase laterals.

Reclosers

Device	Location	Relay / curve	Pickup	TDS fast / delayed	Source of setting
R1	RG60 end of line 650–632	GE IAC77B801A, extremely inverse	720 A (CT 900/5, tap 4 A)	0.5 / 10	Paper ($I_p = 1.25 \times 587.6$ A)
R2 forward	671 end of line 632–671	GE IAC77B801A, extremely inverse	600 A (CT 1000/5, tap 3 A)	0.8 / 4.0	Fitted to Table III (not given in the paper)
R2 reverse (DSDR)	same	GE IAC extremely inverse	322 A	0.276 / 4.582	Eq. (12): 1.25×257.8 A reverse load current with DG; TDS by step 8 of the method

Fuses

15 fuses placed as in Fig. 6 of the paper (19 objects: the distributed load is split into five sections in the model, each with an F-DL fuse). Each fuse has its own fuse type: minimum melting time from eq. (6), $\log_{10}(t) = -1.8 \log_{10}(I) + b_i$ with the paper's Table III b_i ; total clearing at $1.21 \times$ the melting current (band width of a tested DlgSILENT library fuse, Gould-Shawmut A055B, used as the template).

Fuse	Location	b_i	Fuse	Location	b_i
F632	632 end of 632–645	6.38	F692	load at 692	6.13
F633	632 end of 632–633	6.83	F692-R	692 end of 692–675	6.14
F634	XFM-1 LV side	8.32	F675	load at 675	6.19
F645	load at 645	6.65	F671-2	671 end of 671–684	5.87
F646	645 end of 645–646	6.67	F684	684 end of 684–611	6.15
F-DL	distributed load	6.54	F611	load at 611	6.19
F671	671 end of 671–680	6.08	F652	684 end of 684–652	6.17
F671-1	671 end of switch 671–692	5.85			

2. Table II – maximum, minimum and rated branch currents

Rated current I_{nom} : unbalanced load flow, largest phase current at the device end. $I_{f,max}$: bolted three-phase fault at the downstream node, maximum short-circuit currents; at 645, 646 and 684 (two phases) and 611, 652 (one phase) a three-phase fault cannot exist, so the largest fault that can (line-line or line-to-ground) is used. $I_{f,min}$: single line-to-ground fault through 3 ohm at the farthest node, minimum short-circuit currents. DG out of service. Match = $100\% - |model - paper| / paper$.

From	To	I_{nom} model (A)	paper	match	$I_{f,min}$ model (kA)	paper	match	$I_{f,max}$ model (kA)	paper	match
RG60	632	587.6	587.1	100 %	1.08	1.24	87 %	7.91	5.41	54 %
632	633	81.1	81.1	100 %	0.77	0.72	93 %	6.26	6.73	93 %
632	645	143.4	143.3	100 %	0.84	0.93	90 %	4.67	4.25	90 %
632	671	479.0	478.2	100 %	0.91	1.08	84 %	4.71	4.52	96 %
XFM1-HV	side	81.1	81.1	100 %	0.73	0.73	100 %	6.26	6.48	97 %
XFM1-LV	side	706.3	704.7	100 %	0.70	0.59	81 %	18.20	18.76	97 %
645	646	65.4	64.9	99 %	0.77	0.84	92 %	4.01	3.64	90 %
671	692	229.8	229.3	100 %	0.73	0.78	94 %	4.71	3.69	72 %
671	684	71.1	70.9	100 %	0.66	0.69	96 %	3.39	3.47	98 %
671	680	0.0	0.0	100 %	0.64	0.59	92 %	3.86	6.25	62 %
692	675	205.9	205.9	100 %	0.73	0.71	97 %	4.25	7.83	54 %
684	611	71.1	70.8	100 %	0.68	0.63	92 %	2.20	1.93	86 %
684	652	63.0	63.1	100 %	0.66	0.66	100 %	2.20	2.67	82 %
Average				100 %			92 %			82 %

Rated currents match the paper within 1 % on every branch, and $I_{f,min}$ within 20 %. $I_{f,max}$ differs by more than 20 % on four branches: RG60–632 (model 46 % higher), 671–692 (28 % higher), and 671–680 and 692–675 (model 38 % and 46 % lower). The paper's 6.25 and 7.83 kA at 680 and 675 exceed its own 5.41 kA for the feeder-head branch RG60–632, which a radial feeder without DG cannot produce; the paper does not state the source impedance or DG state it used.

3. Table III – fuse coefficients b_i

$b_i = \log_{10}(t_F + i/(z+1) \cdot (t_D - t_F)) + 1.8 \cdot \log_{10}(I_F)$ (eqs. 7, 9), with the recloser fast and delayed times at a bolted fault below each fuse.

Fuse	Recloser	i/z	I_{fuse} (A)	t_F (s)	t_D (s)	b_i model	b_i paper	difference
F632	R1	1 / 2	4668	0.081	1.625	6.38	6.38	+0.00
F633	R1	1 / 2	6258	0.056	1.117	6.45	6.83	-0.38
F634	R1	2 / 2	18198	0.261	5.227	8.22	8.32	-0.10
F645	R1	2 / 2	4668	0.081	1.625	6.65	6.65	+0.00
F646	R1	2 / 2	4007	0.099	1.983	6.62	6.67	-0.05
F-DL	R1	1 / 1	4707	0.084	1.673	6.55	6.54	+0.01
F671	R2	1 / 1	3865	0.131	0.653	6.05	6.08	-0.03
F671-1	R2	1 / 3	4705	0.101	0.507	5.92	5.85	+0.07
F692	R2	2 / 2	4705	0.101	0.507	6.18	6.13	+0.05
F692-R	R2	2 / 3	4247	0.116	0.578	6.07	6.14	-0.07
F675	R2	3 / 3	4247	0.116	0.578	6.20	6.19	+0.01
F671-2	R2	1 / 3	3389	0.151	0.753	5.83	5.87	-0.04
F684	R2	2 / 3	2198	0.290	1.452	5.96	6.15	-0.19
F611	R2	3 / 3	2198	0.290	1.452	6.08	6.19	-0.11
F652	R2	2 / 2	2199	0.282	1.409	6.03	6.17	-0.14

11 of 15 b_i within 0.1 of the paper; F633 (−0.38), F684 (−0.19), F652 (−0.14) and F611 (−0.11) differ most. The fuses in the model use the paper's b_i .

4. Table IV – operating times for a fault at each node

Largest bolted fault that exists at the node, DG in service. R2 uses its reverse (DSDR) setting for nodes upstream of R2. Fuse: melting time of the primary fuse in the faulted path. Times in seconds; model on the left, paper on the right. Fault currents from the fault study (study-case short-circuit setting: minimum currents).

Node	Fault	Model						Paper				
		R1 fast	R1 del.	R2 fast	R2 del.	Fuse	Fuse MMT	R1 fast	R1 del.	R2 fast	R2 del.	Fuse MMT
632	LLL	0.046	0.912	0.058	0.958	---	---	0.070	1.398	0.045	0.514	---
633	LLL	0.065	1.291	0.089	1.472	F633	0.835	0.094	1.880	0.052	0.706	0.163
645	LLG	0.090	1.795	0.124	2.063	F632	0.478	0.105	2.095	0.087	1.659	0.276
646	LLG	0.113	2.255	0.162	2.697	F646	1.248	0.128	2.555	0.103	2.046	0.181
DL	LLL	0.065	1.309	0.051	0.841	F-DL	0.369	0.087	1.733	0.076	1.402	0.248
671	LLL	0.089	1.775	0.111	0.553	---	---	0.105	2.091	0.068	1.113	---
692	LLL	0.089	1.775	0.111	0.553	F671-1	0.194	0.115	2.307	0.072	1.258	0.175
675	LLL	0.106	2.128	0.134	0.669	F692-R	0.233	0.121	2.414	0.074	1.332	0.185
684	LLG	0.140	2.803	0.170	0.852	F671-2	0.189	0.125	2.507	0.075	1.394	0.205
680	LLL	0.130	2.590	0.159	0.795	F671	0.261	0.137	2.747	0.079	1.555	---
611	LG	0.350	6.998	0.434	2.171	F684	1.187	0.326	6.523	0.146	4.385	0.417
652	LG	0.375	7.495	0.435	2.175	F652	1.305	0.676	13.525	0.143	9.370	0.233

- R1: the model's delayed/fast ratio is 20 at every node, exactly as in the paper (TDS 0.5 / 10 on one curve). Absolute R1 times agree within 5–45 % (closest at 680, 684 and 611); the paper does not state the fault type used for Table IV.
- R2: the paper's delayed/fast ratio varies from 11 to 65 between nodes, so its fast and delayed stages must use different curves or pickups. These settings are not published; the model's R2 uses one curve (ratio 5 forward).
- Fuses: the model's melting times are 0.9–7 times the paper's, because the fuses follow eq. (6) with the Table III b_i (see section 7).

Recloser R1 at the paper's own operating points

Paper source	Device	Current (A)	Paper (s)	Model (s)	Model / paper
Text (F645 example)	R1 fast	4219.0	0.097	0.096	0.99
Text (F645 example)	R1 delayed	4219.0	1.932	1.926	1.00
Fig 11 LL 646, 1 ohm	R1 fast	3492.2	0.134	0.129	0.96
Fig 11 LL 646, 1 ohm	R1 delayed	3492.2	2.690	2.579	0.96
Fig 16 LL 646	R1 fast	3605.9	0.128	0.123	0.96
Fig 16 LL 646	R1 delayed	3605.9	2.555	2.451	0.96

5. Figs. 14 and 17 – recloser–fuse coordination maps

Coordination holds when every recloser feeding the fault trips on its fast curve before the primary fuse melts, and the fuse clears before the recloser's delayed trip. Upstream of R2 both R1 (grid) and R2 (DG, reverse direction) feed the fault. Node 634 (fuse on the transformer LV side) is not coordinated, as in the paper; 632 and 671 have no fuse in the faulted path.

Fig. 14 (model): recloser–fuse coordination without the DSDR

Conventional R2 (one setting for both directions); DG in service
✓ held 20 ✗ lost 11 – not applicable (no fuse in the path or fault type absent)

Fault type	632	633	634	645	646	671	692	675	680	684	652	611	DL
LG	–	✗	–	✓	✓	–	✓	✓	✓	✓	✓	✓	✓
LL	–	✗	–	✗	✗	–	✓	✓	✓	✓	–	–	✗
LLG	–	✗	–	✗	✗	–	✓	✓	✓	✓	–	–	✗
LLL	–	✗	–	–	–	–	✓	✓	✓	–	–	–	✗

Model vs paper, cells rated in both: **23 agree, 8 differ** (633 LG, LL; 645 LG; 675 LLG, LLL; DL LL, LLG, LLL).

Fig. 17 (model): recloser–fuse coordination with the DSDR

R2 with separate reverse settings (Ip 322 A, TDS 0.28 / 4.58)
✓ held 29 ✗ lost 2 – not applicable (no fuse in the path or fault type absent)

Fault type	632	633	634	645	646	671	692	675	680	684	652	611	DL
LG	–	✗	–	✓	✓	–	✓	✓	✓	✓	✓	✓	✓
LL	–	✗	–	✓	✓	–	✓	✓	✓	✓	–	–	✓
LLG	–	✓	–	✓	✓	–	✓	✓	✓	✓	–	–	✓
LLL	–	✓	–	–	–	–	✓	✓	✓	–	–	–	✓

Model vs paper: **29 agree, 2 differ** (633 LG and LL: R1's delayed trip comes before F633 clears, a consequence of the paper's $b_1 = 6.83$ for F633).

With the DSDR, R2's separate reverse setting restores coordination at 645, 646 and DL for all fault types and at 633 for LLG and LLL, which is the paper's central result.

6. Operation sequences (time–current plots)

Figs. 8, 9, 11, 12, 13 and 16 are drawn by PowerFactory itself (Figure_Plots.py): each page has the figure's reclosers and fuses, and the figure's fault is applied with the DG in service, so the vertical lines are PowerFactory's fault currents and the labels its trip times. At the fault study's currents these labels equal the model's calculated times exactly (Fig. 16: R1 0.129 / 2.572 s, F646 1.406 s, F632 0.712 s), confirming that the fuse curves in PowerFactory follow eq. (6). The plots below use maximum short-circuit currents, so their currents are 5–10 % higher and their times shorter than in Table IV (minimum currents). Fig. 15 needs R2's reverse (DSDR) setting, which is not yet a relay in the model, and Fig. 7 is not a time-current plot; both are calculated in Python from PowerFactory's fault currents.

Fig. 8 (PowerFactory model) - SLG fault at node 611, DG in service

R2: blue = fast, red = delayed | F684: green band | F671-2: cyan band (fuse band: melting to clearing time)
Fault current: R2 2037 A, F684 2507 A, F671-2 2507 A | Paper: R2 fast 0.207 s, F684 0.476 s, F671-2 1.456 s, R2 delayed 2.472 s

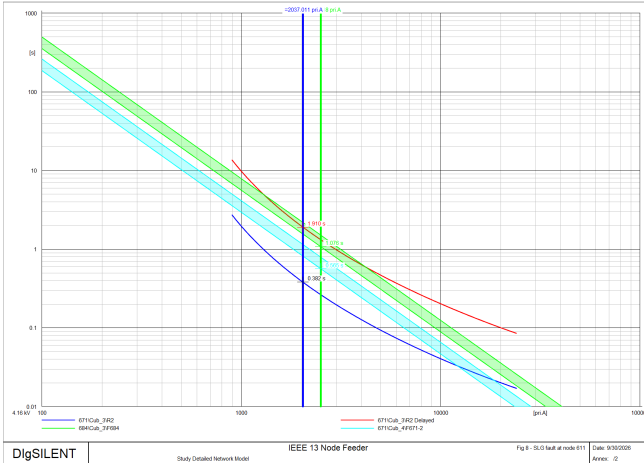


Fig. 8 (PowerFactory). SLG fault at 611. Paper: R2 fast 0.207 s, F684 0.476 s, F671-2 1.456 s, R2 delayed 2.472 s.

Fig. 11 (PowerFactory model) - LL fault at node 646, Rf 1 ohm, DG in service

R1: blue = fast, red = delayed | F646: green band | F632: cyan band (fuse band: melting to clearing time)
Fault current: R1 2269 A, F646 2585 A, F632 2624 A | Paper: R1 fast 0.134 s, F646 0.202 s, F632 1.096 s, R1 delayed 2.690 s

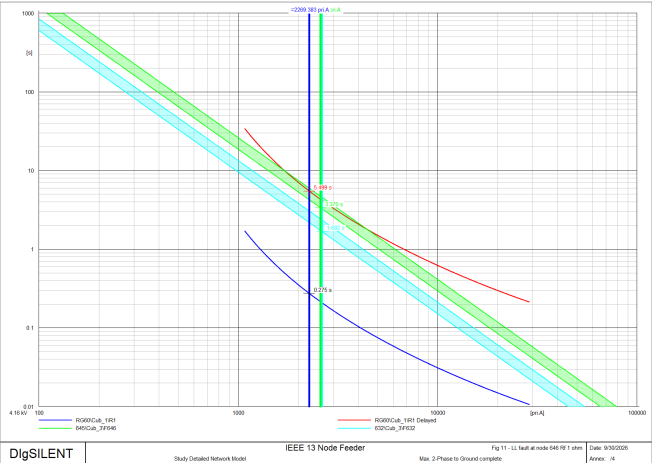


Fig. 11 (PowerFactory). LL fault at 646, Rf 1 ohm. Paper: R1 fast 0.134 s, F646 0.202 s, F632 1.096 s, R1 delayed 2.690 s.

Fig. 13 (PowerFactory model) - 3-phase fault at 10 % of line 632-633, DG in service

R2: blue = fast, red = delayed | F633: green band (fuse band: melting to clearing time)
Fault current: R2 1736 A, F633 9377 A | Paper: R2 fast 0.079 s, R2 delayed 1.556 s, F633 0.022 s

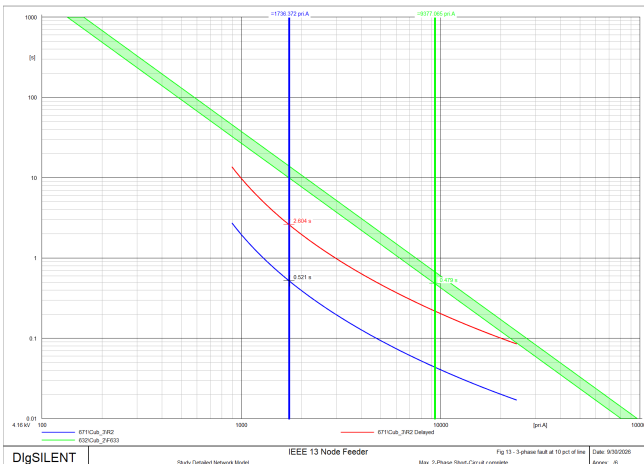


Fig. 13 (PowerFactory). 3-phase fault at 10 % of 632–633. Paper: R2 fast 0.079 s, delayed 1.556 s, F633 0.022 s.

Fig. 16 (PowerFactory model) - solid LL fault at node 646, DG in service

R1: blue = fast, red = delayed | F646: green band | F632: cyan band (fuse band: melting to clearing time)
Fault current: R1 3818 A, F646 4548 A, F632 4579 A | Paper: R1 fast 0.128 s, F646 0.181 s, F632 0.411 s, R1 delayed 2.555 s

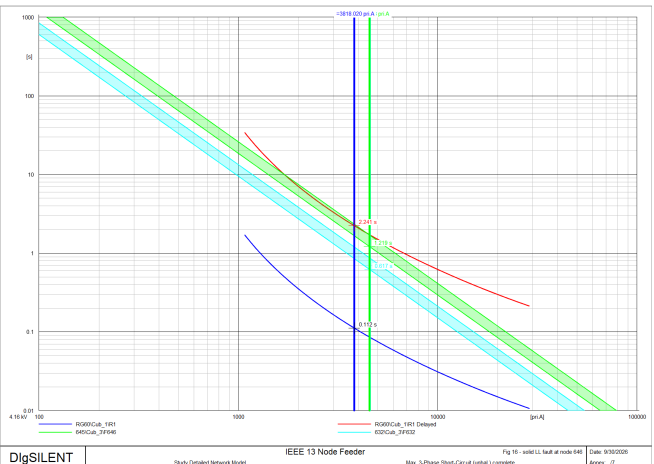


Fig. 16 (PowerFactory). Solid LL fault at 646. Paper: R1 fast 0.128 s, F646 0.181 s, F632 0.411 s, R1 delayed 2.555 s.

Fig. 12 (PowerFactory model) - LL fault at node 645, Rf 1.5 ohm, DG in service
R2: blue = fast, red = delayed | F646: green band | F632: cyan band (fuse band: melting to clearing time)
Fault current: R2 644 A, F646 50 A, F632 2253 A | Paper: R2 fast 0.419 s, R2 delayed 16.683 s, F632 25.440 s

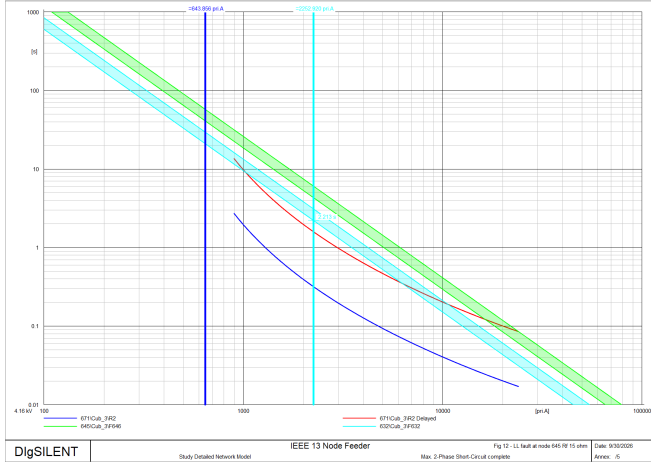


Fig. 12 (PowerFactory). LL fault at 645, Rf 1.5 ohm, conventional R2.
Paper: R2 fast 0.419 s, delayed 16.683 s, F632 25.440 s.

Fig. 9 (PowerFactory model) - LLG fault at mid 692-675, Rf 1 ohm, DG in service
R2: blue = fast, red = delayed | F692-R: green band | F671-1: cyan band (fuse band: melting to clearing time)
Fault current: R2 1600 A, F692-R 2234 A, F671-1 1413 A | Paper: R2 fast 0.141 s, fuses 0.217 s / 0.546 s, R2 delayed 1.483 s

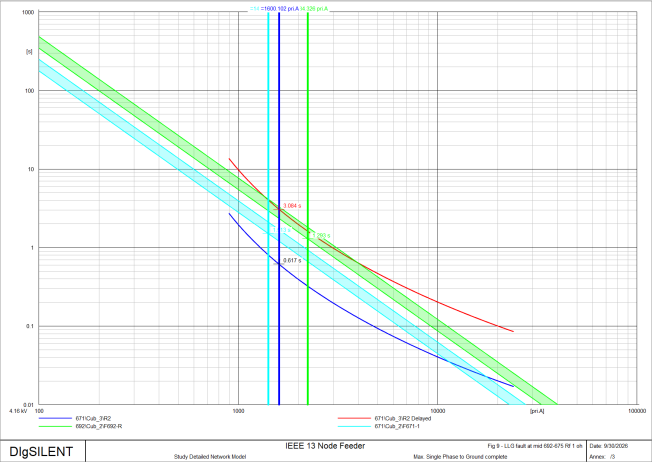


Fig. 9 (PowerFactory). LLG fault at mid 692-675, Rf 1 ohm. Paper:
R2 fast 0.141 s, fuses 0.217 / 0.546 s, R2 delayed 1.483 s.

Fig. 15 (model): solid LL fault at node 646, R2 as DSDR

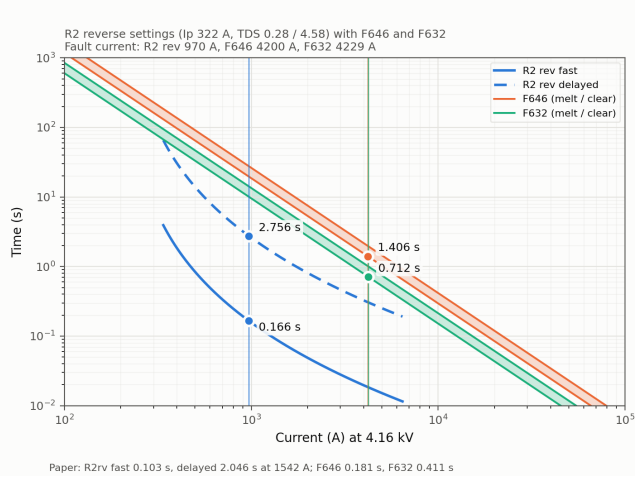


Fig. 7 (model): change in recloser-fuse CTI with DG penetration

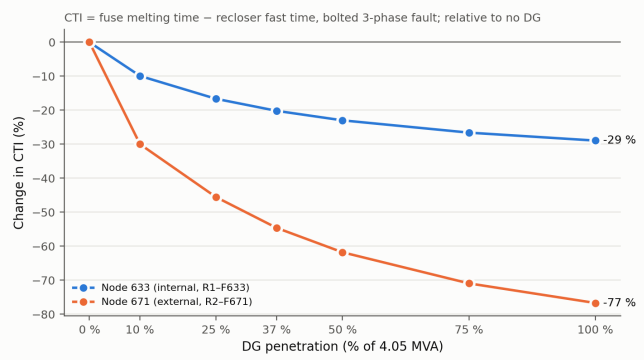


Fig. 7: the recloser-fuse CTI shrinks as the DG grows, by 29 % at node 633 and 77 % at node 671 at full DG rating (penetration taken as % of the 4.05 MVA rating; the paper does not define it).

7. Differences from the paper and their causes

Paper source	Device	Current (A)	Paper (s)	Model (s)	Model / paper
Fig 8 SLG 611	R2 fast	1975.6	0.207	0.405	1.95
Fig 8 SLG 611	F684 melt	1975.6	0.476	1.651	3.47
Fig 8 SLG 611	F671-2 clear	1975.6	1.456	1.221	0.84
Fig 8 SLG 611	R2 delayed	1975.6	2.472	2.023	0.82
Fig 11 LL 646, 1 ohm	F646 melt	3492.2	0.202	1.961	9.71
Fig 11 LL 646, 1 ohm	F632 melt	3492.2	1.096	1.006	0.92
Fig 13 3ph 10% 632-633	R2 fast	3361.4	0.079	0.165	2.09
Fig 13 3ph 10% 632-633	R2 delayed	3361.4	1.556	0.824	0.53
Fig 13 3ph 10% 632-633	F633 melt	8517.2	0.022	0.569	25.88
Fig 15 LL 646 (DSDR)	R2rv fast	1542.1	0.103	0.667	6.48
Fig 15 LL 646 (DSDR)	R2rv delayed	1542.1	2.046	3.337	1.63
Fig 15/16 LL 646	F646 melt	3640.0	0.181	1.820	10.05
Fig 15/16 LL 646	F632 melt	3640.0	0.411	0.933	2.27

- **Fuse curves.** Working back from the paper's figures, its fuses are faster than its Table III b_i allow (e.g. F646: 5.67 from the figures against 6.67 in Table III; F633: 5.42 against 6.83), and they are drawn as curved manufacturer-style bands, not the straight lines of eq. (6). The paper's PowerFactory figures most likely used library fuse types. The model follows eq. (6) and Table III.
- **Series fuses.** With the Table III b_i , the upstream fuse melts before the downstream fuse in all four series pairs (F632–F646, F671-1–F692-R, F671-2–F684, F671-2–F652), which violates the paper's own 75 % rule, eq. (8). The paper's figures show the correct order.
- **R2 settings.** The paper names the relay (GE Alstom CDG34) but gives no pickup or TDS for R2, forward or reverse. The model's forward setting is fitted to Table III; the reverse (DSDR) setting is derived with eq. (12) and step 8 of the paper's method.
- **Fault levels.** The paper's $I_{f,max}$ at 680 and 675 (6.25, 7.83 kA) exceed its own 5.41 kA at the feeder head (RG60–632); the source impedance and DG state used for Table II are not given (section 2).

8. Next steps

- Fig. 10: time-domain (RMS/EMT) simulation of two fast reclosing attempts of R2 before the fuse clears an LL fault at 684.
- Fig. 6 and Table I: redraw the single-line diagram with the protective devices and check the DG data in the model against Table I.
- Discuss with the supervisor whether to keep the eq. (6) fuses or fit fuse curves to the paper's figures (reproduces the plots, but departs from Table III).
- Optional: IEEE 34-node system (Figs. 18–21, Table V).

How to reproduce: in PowerFactory run **Show_Progress.py** (Tables II–IV, Figs. 14 and 17 in the Output Window, saved to Result\Show_Progress_output.txt); for the plots run Fault_Study.py in PowerFactory, then make_figures.py in Python.